

Md Raihanul Haque Rahi

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SUMMARY

Final-year Mechatronics Engineering student with production experience building AI-powered backend systems, real-time voice agent pipelines, and on-device ML applications. Published researcher (Elsevier, 2025); competitive programmer (Codeforces 1280); open-source contributor to a 40k-star Microsoft repository.

EDUCATION

Rajshahi University of Engineering & Technology (RUET)
B.Sc. in Mechatronics Engineering

Rajshahi, Bangladesh
Mar 2022 – Aug 2026 (Expected)

TECHNICAL SKILLS

Languages: Python, C/C++, Dart, Rust, Swift, SQL
Frameworks & Libraries: FastAPI, Django, Flutter, PyTorch, TensorFlow, MLX, LangChain, HuggingFace
Databases & Infra: PostgreSQL, Supabase, Redis, Firebase, Docker, WebRTC, LiveKit, FastRTC
AI / ML: LLM integration (OpenAI, Gemini, Groq), ASR/TTS/VAD pipelines, Computer Vision (YOLO, stereo depth), on-device inference (TFLite, ONNX, MLX), model quantization & optimization

EXPERIENCE

Intelsense AI
Backend Engineer

Apr 2024 – Present
Remote

- Own the backend architecture for an AI-powered SaaS platform, designing **Django/PostgreSQL** systems with a focus on modularity, reliability, and scale
- Integrated production **LLM, ASR, and TTS** inference pipelines into the web backend, enabling real-time AI voice and text interaction features at the product layer

Cortex Technologies
Co-Founder, Edge AI Engineer

Jan 2024 – Present
Self-Employed

- Lead computer-vision model development for client projects: custom object detection pipelines, stereo depth estimation, and real-time vision inference on resource-constrained edge hardware
- Architect language-model integrations for NLP automation tasks, covering model selection, quantization (TFLite/ONNX), and end-to-end deployment on edge devices

Outlier
AI Trainer, Quality Reviewer

Sep 2024 – Dec 2024
Remote

- Evaluated and corrected AI-generated solutions across mathematics and code, contributing to measurable improvement in model reasoning accuracy on complex problem categories

Remotasks
LLM Dataset Inspector

Jul 2023 – Jan 2024
Remote

- Audited LLM training datasets spanning multiple programming languages, enforcing data quality standards that directly impact code-generation fine-tuning pipelines

PROJECTS

Gesprek | FastAPI, PostgreSQL, Redis, Supabase, Docker, ElevenLabs, LiveKit

Co-founded Startup

- Co-founded and shipped a **production voice-agent platform** that enables configurable, niche-specific multi-agent deployments with a modular FastAPI backend, Supabase auth, and Redis-backed real-time TTS
- Engineered multi-tenant agent orchestration, subscription management, and independently scalable containerized services; platform is live and onboarding its first clients

StereoLite: Lightweight Stereo Depth Estimation | *PyTorch, ONNX · B.Sc. Thesis*

Active

- Designed an **11.1M-parameter** stereo depth model that is **3× smaller** than DEFOM-Stereo (30M+ params) and requires **4× less VRAM** (1.5 GB vs. 6 GB), achieving EPE 0.57 in 1 600 training steps
- Decoupled iterative multi-scale GRU refinement from a frozen DINOv2 backbone, demonstrating that refinement quality outperforms backbone scale at a fraction of compute and memory cost

Hishab: Agentic Expense Tracker | *Flutter, FastAPI, Google Gemini, Supabase*

May 2024 – Aug 2025

- Built a production Flutter + FastAPI app that turns Banglish (Bengali/English) natural-language prompts into structured financial records using **Google Gemini function-calling**
- Implemented multi-tenant auth with Supabase row-level security, SMTP notification flows, and a companion web client

Diet Engine: AI Nutrition Coach | *Flutter, FastAPI, PyTorch, Firebase*

Feb – May 2024

- **Published** peer-reviewed research in *Food Chemistry Advances* (Elsevier, 2025) on a real-time food recognition and personalised nutrition system; doi:10.1016/j.focha.2025.100978
- Designed the complete stack: Flutter mobile client, FastAPI inference server with PyTorch food classification models, and Firebase authentication

Perceptron: On-Device Image Classifier | *Flutter, TFLite*

Nov 2025

- Designed a **model-agnostic** TFLite inference engine in Flutter with an error-adaptive backend-swap architecture, enabling any classification model to run entirely on-device without server dependency

OPEN SOURCE

Microsoft / VibeVoice | *40,000+ GitHub stars, Emotional Voice Synthesis*

2024

- Contributed package-import support to a state-of-the-art emotional TTS model maintained under the Microsoft organisation

ACHIEVEMENTS & ACTIVITIES

- **JAXA Kibo Robot Programming Challenge 2025: 1st place, Bangladesh regional; 4th place globally;** programmed Astrobees robots for zero-gravity navigation and AR object recognition aboard the ISS (improved from 2nd place in 2024)
- **Competitive Programming:** Codeforces rating **1280** (Pupil); data structures, algorithms, combinatorics
- **National Robotics Competition, RUET 2022 (Team Cortex):** Designed and built a fully autonomous vision-guided 6-DOF robotic arm
- **Coding Club Lead:** Led RUET student coding club; taught C/C++ and algorithmic problem-solving to first-year engineering students
- **Rahi 404 (YouTube):** Runs a technical channel covering Python, machine learning, and AI engineering.